Rudhira Metal Industries

ISO 9001-2015 CERTIFIED COMPANY

OFFICIAL TECHNICAL SPECIFICATION CATALOGUE

FLANGES DIMENSIONS CATALOGUE

DIMENSIONAL DATA & ENGINEERING STANDARDS

Complete Technical Specifications, Dimensions, Weights & Tolerances

COVERED TECHNICAL STANDARDS & SPECIFICATIONS

- ASME B16.5 (2020) — Class 150#, 300#, 600#, 900#, 1500# & 2500# Flanges
- Weld Neck, Slip-On, Blind, Socket Weld, Threaded & Lapped Flanges (1/2" to 24" NB)
- Welding Neck Flange Bores (B) — Sch 20, 30, 40S, 40, XS, 80, 120, 160, XXS
- ANSI Flange Estimated Weights (Kgs) for Weld Neck & Slip-On Flanges
- EN 1092-1 / DIN Standard — Dimensions of PN 10 Flanges (DN 10 to DN 800)
- EN 1092-1 / DIN Standard — Dimensions of PN 16 Flanges (DN 10 to DN 800)
- BS-10 Standard — Plate Flanges Table D, Table E, Table F, Table H
- ASME B16.5 General Dimensional Tolerances (Facings, Hubs, Bores, Drilling)

RUDHIRA METAL INDUSTRIES — QUALITY ASSURANCE

ISO 9001:2015 Certified Manufacturing & Stockholding Facility

Material Test Certificates (EN 10204 3.1) • Positive Material Identification (PMI)

Third Party Inspection: Bureau Veritas, DNV, TUV, Lloyd's Register, PDIL, SGS

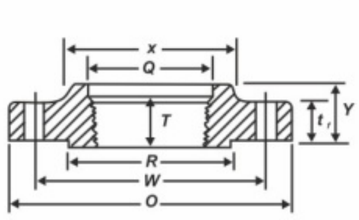
RUDHIRA METAL INDUSTRIES

Commercial Office: Khetwadi, Mumbai - 400 004, Maharashtra, India

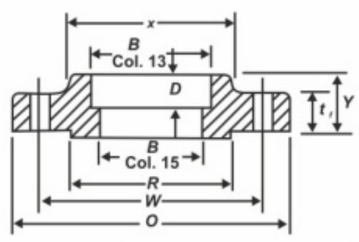
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Phone: +91 98920 71909 / +91 99879 73244 • Email: sales@rudhirametal.com • Web: www.rudhirametal.com

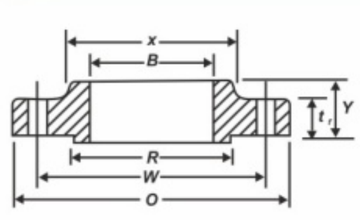
DIMENSIONS OF CLASS 150 FLANGES



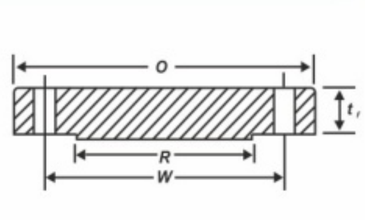
Threaded



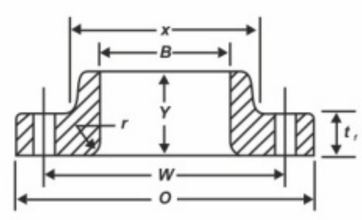
Socket Welding (½" to 2½" Only)



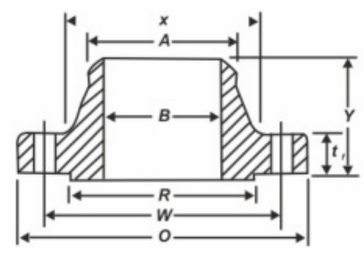
Slip-On Welding



Blind



Lapped



Welding Neck

1	2	3	4	5	6	7	9	10	11	12	13	14	15	16	17	18	19	20
Nominal Pipe Size NPS	Outside Diameter of Flange	Thickness of Flange, Min., t_f	Thick-ness Lap Joint Min., t_f	Drilling			Diameter of Hub, X	Hub Diameter Begin-ning of Chamfer Welding Neck A	Length Thru Hub			Threaded/ Length Threaded Min, T	Bore			Corner Radius of Bore of Lapped Flange and Pipe, R	Depth of Socket, D	Dia-meter of RF R
				Diameter of Bolt Circle W	Diameter of Bolt Holes	Number of Bolts			Threaded/ Slip-On/ Socket Welding, Y	Lapped, Y	Welding Neck, Y		Slip-On/ Socket Welding, Min. B	Lapped Min., B	Welding Neck/ Socket Welding B [Note (2)]			
1/2	89	9.7	11.2	60.3	15.9	4	30	21.3	14	16	46	16	22.2	22.9	15.8	3	10	34.9
3/4	99	11.2	12.7	69.8	15.9	4	38	26.7	14	16	51	16	27.7	28.2	20.9	3	11	42.9
1	108	12.7	14.3	79.4	15.9	4	49	33.4	16	17	54	17	34.5	34.9	26.6	3	13	50.8
1 1/4	117	14.3	15.9	88.9	15.9	4	59	42.2	19	21	56	21	43.2	43.7	35.1	5	14	63.5
1 1/2	127	15.9	17.5	98.4	15.9	4	65	48.3	21	22	60	22	49.5	50.0	40.9	6	16	73.0
2	152	17.5	19.1	120.7	19.1	4	78	60.3	24	25	62	25	61.9	62.5	52.5	8	17	92.1
2 1/2	178	20.7	22.3	139.7	19.1	4	90	73.0	27	29	68	29	74.6	75.4	62.7	8	19	104.8
3	190	22.3	23.9	152.4	19.1	4	108	88.9	29	30	68	30	90.7	91.4	77.9	10	21	127.0
3 1/2	216	22.3	23.9	177.8	19.1	8	122	101.6	30	32	70	32	103.4	104.1	90.1	10	....	139.7
4	229	22.3	23.9	190.5	19.1	8	135	114.3	32	33	75	33	116.8	116.8	102.3	11	....	157.2
5	254	22.3	23.9	215.9	22.3	8	164	141.3	35	36	87	36	143.8	144.4	128.2	11	....	185.7
6	279	23.9	25.4	241.3	22.3	8	192	168.3	38	40	87	40	170.7	171.4	154.1	13	....	215.9
8	343	27.0	28.6	298.2	22.3	8	246	219.1	43	44	100	44	221.5	222.2	202.7	13	....	269.9
10	406	28.6	30.2	362.0	25.4	12	3.5	273.0	48	49	100	49	276.2	277.4	254.6	13	....	323.8
12	483	30.2	31.8	431.8	25.4	12	365	323.8	54	56	113	56	327.0	328.2	304.8	13	....	381.0
14	533	33.4	35.0	476.3	28.6	12	400	355.6	56	79	125	57	359.2	360.2	To be Specified by Purchaser	13	....	412.8
16	597	35.0	36.6	539.8	28.6	16	457	406.4	62	87	125	64	410.5	411.2		13	....	469.9
18	635	38.1	39.7	577.9	31.8	16	505	457.0	67	97	138	68	461.8	462.3		13	....	533.4
20	698	41.1	42.9	635.0	31.8	20	559	508.0	71	103	143	73	513.1	514.4		13	....	584.2
24	813	46.1	47.7	749.3	35.0	20	663	610.0	81	111	151	83	616.0	616.0		13	....	692.2

DIMENSIONS OF CLASS 300 FLANGES

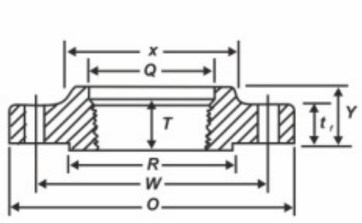
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Nominal Pipe Size NPS	Outside Diameter of Flange o	Thickness of Flange, Min., t,	Thickness Lap Joint Min., t,	Drilling			Diameter of Hub, X	Hub Diameter Begin-ning of Chamfer Welding Neck A	Length Thru Hub			Threaded/ Length Threaded Min, T	Bore			Corner Radius of Bore of Lapped Flange and Pipe, R	Counter-bore Threaded FLANGE Min., Q	Depth of Socket, D	Diameter of RF R
				Diameter of Bolt Circle W	Diameter of Bolt Holes	Number of Bolts			Threaded/ Slip-On/ Socket Welding, Y	Lapped, Y	Welding Neck, Y		Slip-On/ Socket Welding, Min. B	Lapped Min., B	Welding Neck/ Socket Welding B [Note (2)]				
1/2	95	12.7	14.3	66.7	15.9	4	38	21.3	21	22	51	16	22.2	22.9	15.8	3	23.6	10	34.9
3/4	117	14.3	15.9	82.6	19.1	4	48	26.7	24	25	56	16	27.7	28.2	20.9	3	29.0	11	42.9
1	124	15.9	17.5	88.9	19.1	4	54	33.4	25	27	60	18	34.5	34.9	26.6	3	35.8	13	50.8
1 1/4	133	17.5	19.1	98.4	19.1	4	64	42.2	25	27	64	21	43.2	43.7	35.1	5	44.4	14	63.5
1 1/2	155	19.1	20.7	114.3	22.2	4	70	48.3	29	30	67	23	49.5	50.0	40.9	6	50.3	16	73.0
2	165	20.7	22.3	127.0	19.0	8	84	60.3	32	33	68	29	61.9	62.5	52.5	8	63.5	17	92.1
2 1/2	190	23.9	25.4	149.2	22.3	8	100	73.0	37	38	75	32	74.6	75.4	62.7	8	76.2	19	104.8
3	210	27.0	28.6	168.1	22.3	8	117	88.9	41	43	78	32	90.7	91.4	77.9	10	92.2	21	127.0
3 1/2	229	28.6	30.2	184.2	22.3	8	133	101.6	43	44	79	37	103.4	104.1	90.1	10	104.9	....	139.7
4	254	30.2	31.8	200.0	22.3	8	146	114.3	46	48	84	37	116.1	116.8	102.3	11	117.6	....	157.2
5	279	33.4	35.0	235.0	22.3	8	178	141.3	49	51	97	43	143.8	144.4	128.2	11	144.4	....	185.7
6	318	35.0	36.6	269.9	22.3	12	206	168.3	51	52	97	47	170.7	171.4	154.1	13	171.1	....	215.9
8	381	39.7	41.3	330.2	25.4	12	260	219.1	60	62	110	51	221.5	222.2	202.7	13	222.2	....	269.9
10	444	46.1	47.7	387.4	28.6	16	321	273.0	65	95	116	56	276.2	277.4	254.6	13	276.2	....	323.8
12	521	49.3	50.8	450.8	31.8	16	375	323.8	71	102	129	61	327.0	328.2	304.8	13	328.6	....	381.0
14	584	52.4	54.0	514.4	31.8	20	425	355.6	75	111	141	64	359.2	360.2	To be Specified by Purchaser	13	360.4	....	412.8
16	648	55.6	57.2	571.5	35.0	20	483	406.4	81	121	144	69	410.5	411.2		13	411.2	....	469.9
18	711	57.8	60.4	628.6	35.0	24	533	457.0	87	130	157	70	461.8	462.3		13	462.0	....	533.4
20	775	62.0	63.5	685.8	35.0	24	587	508.0	94	140	160	74	513.1	514.4		13	512	....	584.2
24	914	68.3	69.9	812.8	41.3	24	702	610.0	105	152	167	83	616.0	616.0		13	614.4	....	692.2

NOTE : (1) Height of RF 1.5 mm

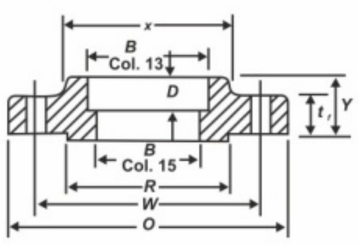
As Per ASME B.16.5 Standard 2020

(2) Dimension in Column 16 Correspond to the inside diameters of pipe as given in ASME B36.10M for Standard Wall pipe, Thickness of Standard Wall is the same as Schedule 40 in sizes NPS 10 and smaller. These bore sizes are furnished unless otherwise specified by the purchaser.

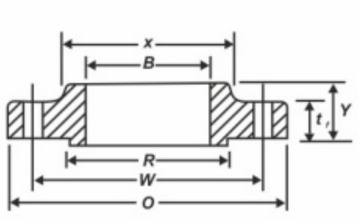
DIMENSIONS OF CLASS 600 FLANGES



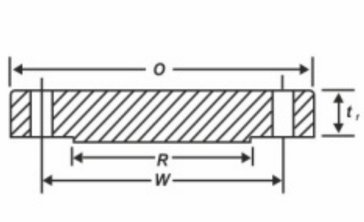
Threaded (NPS 1/2" to 2 1/2" Only



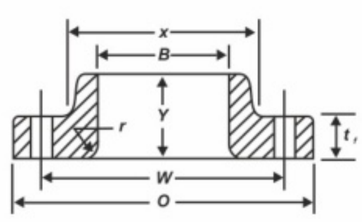
Socket Welding (½" to 2½" Only)



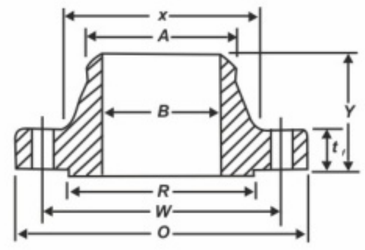
Slip-On Welding



Blind



Lapped



Welding Neck

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Nominal Pipe Size NPS	Outside Diameter of Flange *O*	Thickness of Flange, Min., *t*	Diameter of Hub, *X*	Drilling			Hub Diameter Beginning of Chamfer Welding Neck, *A*	Length Thru Hub			Threaded/ Length Threaded Flange Min., *T*	Bore			Corner Radius of Bore of Lapped Flange and Pipe, *r*	Counter-bore Threaded Flange Min., *Q*	Socket Weld *D*	Diameter of RF *R*
				Diameter of Bolt Circle, *W*	Diameter of Bolt Holes, in.	Number of Bolts		Threaded/ Slip-On, *Y*	Lapped, *Y*	Welding Neck, *Y*		Slip-On Min., *B*	Lapped Min., *B*	Welding Neck, *B*				
½	95	14.3	38	66.5	15.9	4	21.3	22	22	52	16	22.2	22.9	To be Specified by Purchaser	3	23.6	10	34.9
3/4	117	15.9	48	82.6	19.1	4	26.7	25	25	57	16	27.7	28.2		3	29.0	11	42.9
1	124	17.5	54	88.9	19.1	4	33.4	27	27	62	18	34.5	34.9		3	35.8	13	50.8
1 1/4	133	20.7	64	98.6	19.1	4	42.2	29	29	67	21	43.2	43.7		5	44.4	14	63.5
1 ½	155	22.3	70	114.3	22.3	4	48.3	32	32	70	23	49.5	50.0		6	50.6	16	73.0
2	165	25.4	84	127.0	19.1	8	60.3	37	37	73	29	61.9	62.5		8	63.5	17	92.1
2 ½	190	28.6	100	149.2	22.3	8	73.0	41	41	79	32	74.6	75.4		8	76.2	19	104.8
3	210	31.8	117	168.3	22.3	8	88.9	46	46	83	35	90.7	91.4		10	92.2	21	127.0
4	273	38.1	152	215.9	25.4	8	114.3	54	54	102	42	116.8	116.8		11	117.6	---	157.2
5	330	44.5	189	266.7	28.6	8	141.3	60	60	114	48	143.8	144.4		11	144.4	....	185.7
6	356	47.7	222	292.1	28.6	12	168.3	67	67	117	51	170.7	171.4		13	171.4	....	215.9
8	419	55.6	273	349.2	31.8	12	219.1	76	76	133	58	221.5	222.2		13	222.2	....	269.9
10	508	63.5	343	431.8	35.0	16	273.0	86	111	152	66	276.2	277.4		13	276.2	....	323.8
12	559	66.7	400	489.0	35.0	20	323.8	92	117	156	70	327.0	328.2		13	328.6	....	381.0
14	603	69.9	432	527.0	38.1	20	355.6	94	127	165	74	359.2	360.2		13	360.4	....	412.8
16	686	76.2	495	603.2	41.3	20	406.4	106	140	178	78	410.5	411.2		13	411.2	....	469.9
18	743	82.6	546	654.0	44.5	20	457.0	117	152	184	80	461.8	462.3		13	462.0	....	533.4
20	813	88.9	610	723.9	44.5	24	508.0	127	165	190	83	513.1	514.4		13	512.8	....	584.2
24	940	101.6	718	838.2	50.8	24	610.0	140	184	203	93	616.0	616.0		13	614.4	....	692.2

Note : (1) Height of RF 6.4 mm

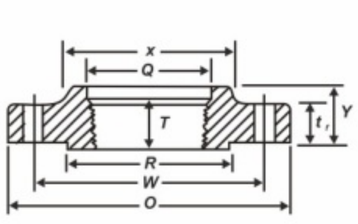
DIMENSIONS OF CLASS 900 FLANGES

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Nominal Pipe Size NPS	Outside Diameter of Flange *O*	Thickness of Flange, Min., *t*	Diameter of Hub, *X*	Drilling			Hub Diameter Beginning of Chamfer Welding Neck, *A*	Length Thru Hub			Threaded/ Length Threaded Flange Min., *T*	Bore			Corner Radius of Bore of Lapped Flange and Pipe, *r*	Counter-bore Threaded Flange Min., *Q*	Socket Weld *D*	Diameter of RF *R*
				Diameter of Bolt Circle, *W*	Diameter of Bolt Holes, in.	Number of Bolts		Threaded/ Slip-On, *Y*	Lapped, *Y*	Welding Neck, *Y*		Slip-On Min., *B*	Lapped Min., *B*	Welding Neck, *B*				
½	121	22.3	38	82.6	22.3	4	21.3	32	32	60	23	22.2	22.9	To be Specified by Purchaser	3	23.6	10	349
3/4	130	25.4	44	88.9	22.3	4	26.7	35	35	70	26	27.7	38.2		3	29.0	11	42.9
1	149	28.6	52	101.6	25.4	4	33.4	41	41	73	29	34.5	34.9		3	35.8	13	50.8
1 1/4	159	28.6	64	111.1	25.4	4	42.2	41	41	73	31	43.2	43.7		5	44.4	14	63.5
1 ½	178	31.8	70	124.0	28.6	4	48.3	44	44	83	32	49.5	50.0		6	50.6	16	73.0
2	216	38.1	105	165.1	25.4	8	60.3	57	57	102	39	61.9	62.5		8	6.5	17	92.1
2 ½	244	41.3	124	190.5	28.6	8	73.0	64	64	105	48	74.6	75.4		8	76.2	19	104.8
3	241	38.1	127	190.5	25.4	8	88.9	54	54	102	42	90.7	91.4		10	92.2	-	127.0
4	292	44.5	159	235.0	31.8	8	114.3	70	70	114	48	116.8	116.8		11	117.6	-	159.0
5	349	50.8	190	279.4	35.0	8	141.3	79	79	127	54	143.8	144.4		11	144.4	-	185.7
6	381	55.6	235	317.5	31.8	12	168.3	86	86	140	58	170.7	171.4		13	171.4	-	215.9
8	470	63.5	298	393.7	38.1	12	219.1	102	114	162	64	221.5	222.2		13	222.2	-	269.9
10	546	69.9	368	469.9	38.1	16	273.0	108	127	184	72	276.2	277.4		13	276.2	-	323.8
12	610	79.2	419	533.4	38.1	20	323.8	117	143	200	77	327.0	328.2		13	328.6	-	381.0
14	641	85.8	451	558.8	41.3	20	355.6	130	156	213	83	359.2	360.2		13	360.4	-	415.8
16	705	88.9	508	616.0	44.5	20	406.4	133	165	216	86	410.5	411.2		13	411.2	-	469.9
18	787	101.6	565	685.8	50.8	20	457.0	152	190	229	89	461.8	462.3		13	462.0	-	533.4
20	857	108.0	622	749.3	54.0	20	508.0	159	210	248	93	513.1	514.4		13	512.8	-	584.2
24	1,041	139.7	749	901.7	66.7	20	610.0	203	267	292	102	616.0	616.0		13	614.4	-	692.2

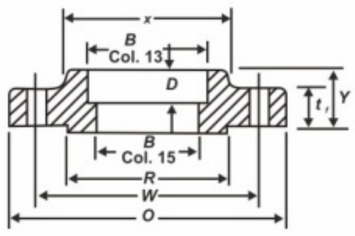
Note : (1) Height of RF 6.4 mm

As Per ASME B.16.5 Standard 2020

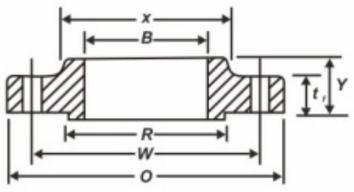
DIMENSIONS OF CLASS 1500 FLANGES



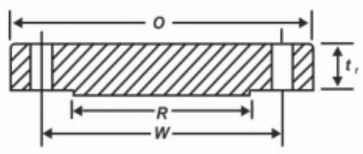
Threaded (NPS 1/2" to 2 1/2" Only)



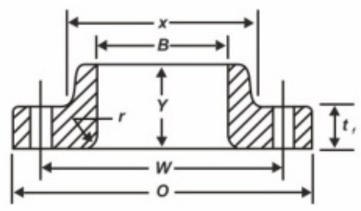
Socket Welding (½" to 2½" Only)



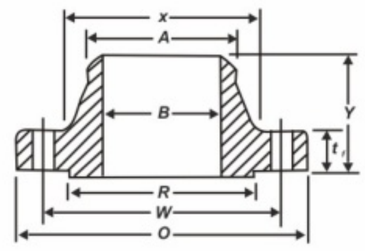
Slip-On Welding (NPS 1/2" to 2 1/2" Only)



Blind



Lapped



Welding Neck

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Nominal Pipe Size NPS	Outside Diameter of Flange *O*	Thickness of Flange, Min., t_f	Diameter of Hub, *X*	Drilling			Hub Diameter Beginning of Chamfer Welding Neck, *A*	Length Thru Hub			Threaded/ Length Threaded Flange Min., *T*	Bore			Corner Radius of Bore of Lapped Flange and Pipe, *r*	Counter-bore Threaded Flange Min., *Q*	Socket Weld *D*	Diameter of RF *R*
				Diameter of Bolt Circle, *W*	Diameter of Bolt Holes, in.	Number of Bolts		Threaded/ Slip-On, *Y*	Lapped, *Y*	Welding Neck, *Y*		Slip-On Min., *B*	Lapped Min., *B*	Welding Neck/ Socket Welding *B*				
½	121	22.3	38	82.6	22.3	4	21.3	32	32	60	23	22.2	22.9	To be Specified by Purchaser	3	23.8	10	34.9
3/4	130	25.4	44	88.9	22.3	4	26.7	35	35	70	26	27.7	28.2		3	29.0	11	42.9
1	149	28.6	52	101.6	25.4	4	33.4	41	41	73	29	34.5	34.9		3	35.8	13	50.8
1 1/4	159	28.6	64	111.1	25.4	4	42.2	41	41	73	31	43.2	43.7		5	44.4	14	63.5
1 ½	178	31.8	70	123.8	28.6	4	48.3	44	44	83	32	49.5	50.0		6	50.6	16	73.0
2	216	38.1	105	165.1	25.4	8	60.3	57	57	102	39	61.9	62.5		8	63.5	17	92.1
2 ½	244	41.1	124	190.5	28.6	8	73.0	64	64	105	48	74.6	75.4		8	76.2	19	104.8
3	267	47.8	133	203.2	31.8	8	88.9	....	73	117	....	....	91.4		10	....	....	127.0
4	311	53.8	162	241.3	35.0	8	114.3	....	90	124	....	....	116.8		11	....	....	157.2
5	375	73.1	197	292.1	41.3	8	141.3	....	105	156	....	....	144.4		11	....	....	185.7
6	394	82.6	229	317.5	38.1	12	168.3	....	119	171	....	....	171.4		13	....	....	215.9
8	483	92.1	292	393.7	44.5	12	219.1	....	143	213	....	....	222.2		13	....	....	269.9
10	584	108.0	368	482.6	50.8	12	273.0	....	178	254	....	....	277.4		13	....	....	323.8
12	673	124.0	451	571.5	54.0	16	323.8	....	219	283	....	....	328.2		13	....	....	381.0
14	749	133.4	495	635.0	60.4	16	355.6	....	241	298	....	....	360.2		13	....	....	412.8
16	826	146.0	552	704.8	66.7	16	406.4	....	260	311	....	....	411.2		13	....	....	469.9
18	914	162.0	597	774.7	73.0	16	457.0	....	276	327	....	....	462.3		13	....	....	533.4
20	984	177.8	641	831.8	79.4	16	508.0	....	292	356	....	....	514.4		13	....	....	584.2
24	1168	203.2	762	990.6	92.1	16	610.0	....	330	406	....	....	616.0		13	....	....	692.2

Note : (1) Height of RF 6.4 mm

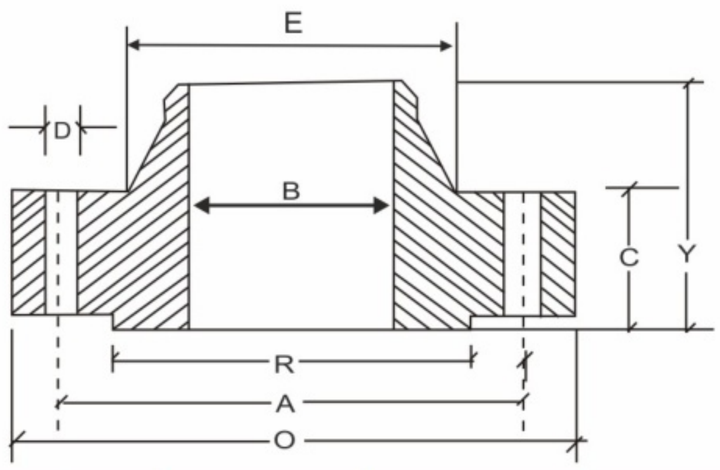
DIMENSIONS OF CLASS 2500 FLANGES

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Nominal Pipe Size NPS	Outside Diameter of Flange *O*	Thickness of Flange, Min., t_f	Diameter of Hub, *X*	Drilling			Hub Diameter Beginning of Chamfer Welding Neck, *A*	Length Thru Hub			Bore			Corner Radius of Bore of Lapped Flange and Pipe, *r*	Counter-bore Threaded Flange Min., *Q*	Diameter of RF *R*
				Diameter of Bolt Circle, *W*	Diameter of Bolt Holes, in.	Number of Bolts		Threaded/ Slip-On, *Y*	Lapped, *Y*	Welding Neck, *Y*	Threaded/ Length Threaded Flange Min., *T*	Lapped Min., *B*	Welding Neck/ Socket Welding, *B*			
½	133	30.2	43	88.9	22.3	4	21.3	40	40	73	29	22.9	To be Specified by Purchaser	3	23.6	34.9
3/4	140	31.8	51	95.2	22.3	4	26.7	43	43	79	32	28.2		3	29.0	42.9
1	159	35.1	57	108.0	25.4	4	33.4	48	48	89	35	34.9		3	35.8	50.8
1 1/4	184	38.1	73	130.2	28.6	4	42.2	52	52	95	39	43.7		5	44.4	63.5
1 ½	203	44.4	79	146.0	31.8	4	48.3	60	60	111	45	50		6	50.6	73.0
2	235	50.99	95	171.4	28.6	8	60.3	70	70	127	51	62.5		8	63.5	92.1
2 ½	267	57.2	114	196.8	31.8	8	73.0	79	79	127	58	75.4		8	76.2	104.8
3	305	66.5	133	228.6	35.0	8	88.9	....	92	168	....	91.4		10	....	127.0
4	356	76.2	165	273.0	41.3	8	114.3	....	106	190	....	116.8		11	....	157.2
5	419	92.1	203	323.8	47.7	8	141.3	....	130	229	....	144.4		11	....	185.7
6	483	108.0	235	368.3	54.0	8	168.3	....	152	273	....	171.4		13	....	215.9
8	552	127.0	305	438.2	54.0	12	219.1	....	178	318	....	222.2		13	....	269.9
10	673	165.1	375	539.8	66.7	12	273.0	....	229	419	....	277.4		13	....	323.8
12	762	184.2	441	619.3	73.0	12	323.8	....	254	464	....	328.2		13	....	381.0

Note : (1) Height of RF 6.4 mm

As Per ASME B.16.5 Standard 2020

WELDING NECK FLANGE BORES (B)



WELDING NECK FLANGE

Nominal Pipe Size	Outside Dia	Sch 20	Sch 30	Std Wall/ Sch 40S	Sch 40	XS	Sch 80	Sch 120	Sch 160	XXS
15	21.33	-	-	15.7	15.7	13.8	13.8	-	11.7	6.4
20	26.67	-	-	20.9	20.9	18.8	18.8	-	15.5	11.0
25	33.40	-	-	26.6	26.6	24.3	24.3	-	20.7	15.2
32	42.2	-	-	35.0	35.0	32.5	32.5	-	29.4	22.8
40	48.3	-	-	40.9	40.9	38.1	38.1	-	34.0	28.0
50	60.3	-	-	52.5	52.5	49.2	49.2	-	42.8	38.1
65	73.0	-	-	62.7	62.7	59.0	59.0	-	53.9	44.9
80	88.9	-	-	77.9	77.9	73.6	73.6	-	66.6	58.4
100	114.3	-	-	102.2	102.2	97.1	97.1	92.0	87.3	80.0
125	141.3	-	-	128.2	128.1	122.2	122.2	115.9	109.5	103.2
150	168.3	-	-	154.0	154.0	146.3	146.3	139.7	131.7	124.3
200	219.1	206.2	204.9	202.7	202.7	193.7	193.7	182.5	173.0	174.6
250	273.0	260.3	257.4	254.0	254.5	247.6	242.8	230.1	251.8	222.2
300	323.85	311.1	307.0	304.8	303.2	298.4	288.8	273.0	257.2	273.0
350	355.6	337.8	336.5	336.5	333.3	330.2	371.5	300.0	284.1	-
400	406.4	390.3	387.3	387.3	381.0	381.0	363.5	344.5	325.4	-
450	457.2	441.1	434.9	438.0	428.4	431.6	409.3	387.1	366.7	-
500	508.0	488.9	482.6	488.9	477.8	482.6	455.6	431.8	407.9	-
600	610.0	590.5	581.0	590.9	575.0	584.2	548.1	518.0	490.9	-

All Fig. for Class 150#

All dimensions are in Millimeters.

ANSI FLANGE WEIGHTS (Kgs.)

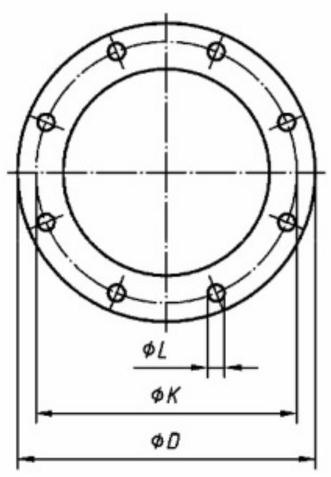
Nominal Pipe size		150 Lb		300 Lb		600 Lb		900 Lb	
N.B.	INCH	WN	SO	WN	SO	WN	SO	WN	SO
15	1/2"	0.49	0.8	0.77	1.2	0.84	1.3	1.88	1.8
20	3/4"	0.73	0.9	1.31	1.4	1.41	1.4	2.51	2.3
25	1"	1.05	0.9	1.64	1.4	1.78	1.8	3.66	3.6
32	1 1/4"	1.36	1.4	2.11	1.8	2.39	2.3	4.25	4.1
40	1 1/2"	1.75	1.4	2.96	2.7	3.38	3.2	5.85	5.5
50	2"	2.62	2.3	3.49	3.2	4.14	4.1	10.9	11.5
65	2 1/2"	4.15	3.2	5.33	4.6	6.1	5.9	14.96	16.5
80	3"	4.99	3.7	7.14	5.9	8.17	7.3	13.34	12
100	4"	7.08	5.9	11.56	10	16.83	16.8	22.03	24
125	5"	8.88	6.8	15.48	12.7	28.4	29	35.38	38
150	6"	10.83	8.6	19.9	17.7	33.88	36	46.84	50
200	8"	17.96	13.7	30.75	26	51.54	52	80.11	78
250	10"	24.56	19.5	44.19	36	85.1	77	118.7	112
300	12"	38.04	29	64.02	52	101.82	91	157.35	148
350	14"	49.98	41	88.93	75	120.58	105	179.97	182
400	16"	62.29	54	110.9	86	169.07	150	216.67	193
450	18"	70.62	59	136.02	113	201.81	182	292.1	258
500	20"	86.76	75	164.55	143	252.51	232	362.87	317
600	24"	118.1	100	238.05	216	354.91	332	665.87	608

NOTE : 1. ALL Weights in Kgs For Standard Weld Neck Flange with RF
2. Different grades of stainless steel have different specific densities and hence their weight may be less than or greater than the weights listed in this table.

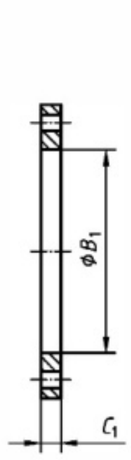
As per Standard ASME B16.5-2020



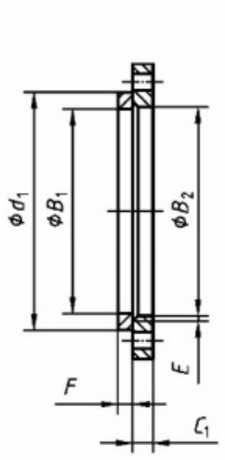
DIMENSIONS OF PN 10 FLANGES



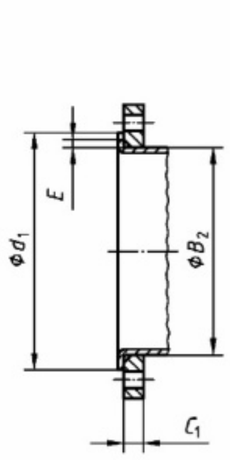
This diagram illustrates the arrangement but not necessarily the correct number of bolt holes. Refer to the column "Bolting Number" for the actual number.



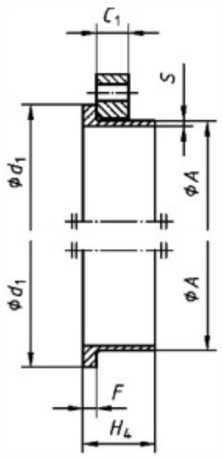
Type 01



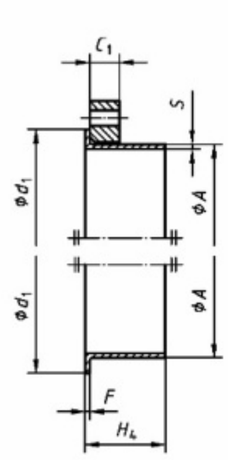
Type 02 and 32



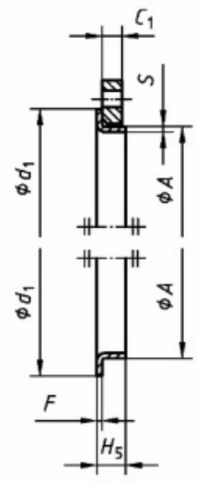
Type 02 and 33



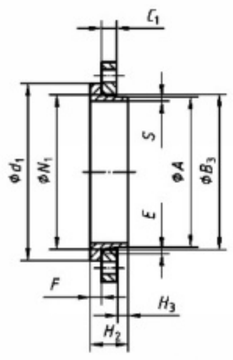
Type 02 and 35



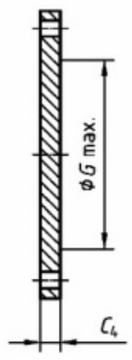
Type 02 and 36



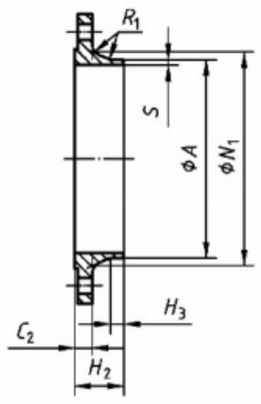
Type 02 and 37



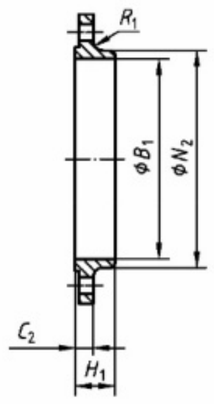
Type 04 and 34



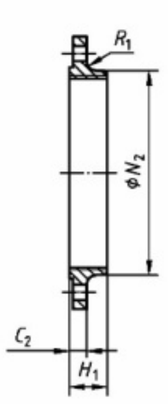
Type 05



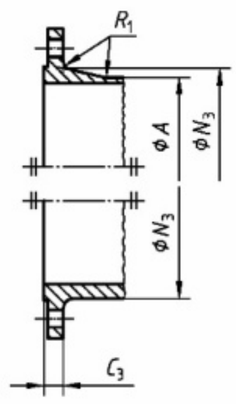
Type 11



Type 12



Type 13



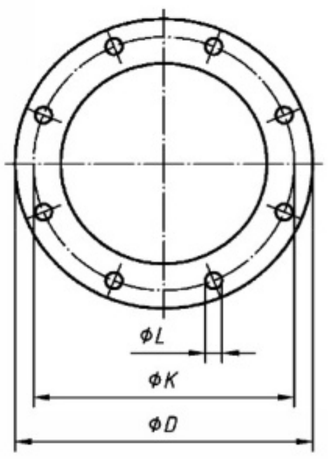
Type 21

DN	Mating dimensions: Outside diameter D	Diameter of bolt circle K	Diameter of bolt hole L	Bolting: Number	Bolting: Size	Outside diameter of neck A	Bore diameters: B_1	Bore diameters: B_2	Bore diameters: B_3	Flange thickness: C_1	Flange thickness: C_2	Flange thickness: C_3	Flange thickness: C_4	Chamfer E	Collar thickness F	Diameter of shoulder G_{max}	Length: H_1	Length: H_2	Length: H_3	Length: H_4	Length: H_5	Neck diameters: N_1	Neck diameters: N_2	Neck diameters: N_3	Corner radii R_1	
	Flange type																									
	01, 02, 04, 05, 11, 12, 13, 21					11 21 [a] 34 [c] 35 - 37	01 12 32	02	04	01 02 04	11 12 13	21	05	02 04	32 34	35	36	37	05	12 13	11 34[c]	11 34[c]	35	36	37	11 34[c]	12 13	21	11 12 13 21, 34	34
10	90	60	14	4	M12	17,2	18,0	21	31	14	16	16	16	3	12	5	2	2,5	—	22	35	6	35	35	7	28	30	28	4	1,8
15	95	65	14	4	M12	21,3	22,0	25	35	14	16	16	16	3	12	5	2	2,5	—	22	38	6	38	38	7	32	35	32	4	2,0
20	105	75	14	4	M12	26,9	27,5	31	42	16	18	18	18	4	14	6	2,5	3	—	26	40	6	40	40	8	40	45	40	4	2,3
25	115	85	14	4	M12	33,7	34,5	38	49	16	18	18	18	4	14	7	2,5	3	—	28	40	6	40	40	10	46	52	50	4	2,6
32	140	100	18	4	M16	42,4	43,5	47	59	18	18	18	18	5	14	8	3	3	—	30	42	6	42	42	12	56	60	60	6	2,6
40	150	110	18	4	M16	48,3	49,5	53	67	18	18	18	18	5	14	8	3	3	—	32	45	7	45	45	15	64	70	70	6	2,6
50	165	125	18	4	M16	60,3	61,5	65	77	20	18	18	18	5	16	8	3	4	—	28	45	8	45	45	20	74	84	84	6	2,9
65	185	145	18	8	M16	76,1	77,5	81	96	20	18	18	18	6	16	8	3	4	55	32	45	10	45	45	20	92	104	104	6	2,9
80	200	160	18	8	M16	88,9	90,5	94	108	20	20	20	20	6	16	10	3	4	70	34	50	10	50	50	25	105	118	120	6	3,2
100	220	180	18	8	M16	114,3	116,0	120	134	22	20	20	20	6	18	10	4	4	90	40	52	12	52	52	25	131	140	140	8	3,6
125	250	210	18	8	M16	139,7	141,5	145	162	22	22	22	22	6	18	10	4	4	115	44	55	12	55	55	25	156	168	170	8	4,0
150	285	240	22	8	M20	168,3	170,5	174	188	24	22	22	22	6	20	10	4	4	140	44	55	12	55	55	25	184	195	190	10	4,5
200	340	295	22	8	M20	219,1	221,5	226	240	24	24	24	24	6	20	11	5	4	190	44	62	16	62	62	30	234	246	246	10	6,3
250	395	350	22	12	M20	273,0	276,5	281	294	26	26	26	26	8	22	12	8	—	235	46	68	16	68	68	—	292	298	298	12	6,3
300	445	400	22	12	M20	323,9	327,5	333	348	26	26	26	26	8	22	12	8	—	285	46	68	16	68	68	—	342	350	348	12	7,1
350	505	460	22	16	M20	355,6	359,5	365	400	30	26	26	26	8	22	13	8	—	330	53	68	16	68	68	—	385	400	408	12	7,1
400	565	515	26	16	M24	406,4	411,0	416	450	32	26	26	26	8	24	14	8	—	380	57	72	16	72	72	—	440	456	456	12	7,1
450	615	565	26	20	M24	457,0	462,0	467	498	36	28	28	28	8	24	15	—	—	425	63	72	16	72	—	—	488	502	502	12	7,1
500	670	620	26	20	M24	508,0	513,5	519	550	38	28	28	28	8	26	16	—	—	475	67	75	16	75	—	—	542	559	559	12	7,1
600	780	725	30	20	M27	610,0	616,5	622	650	42	30	34	34	8	26	18	—	—	575	75	82	18	80	—	—	642	658	658	12	—
700	895	840	30	24	M27	711,0	b	721	—	50	35		38	8	—	20	—	—	670	—	85	18	80	—	—	746	—	772	12	—
800	1 015	950	33	24	M30	813,0		824	—	56	38		48	8	—	20	—	—	770	—	96	18	90	—	—	850	—	876	12	—

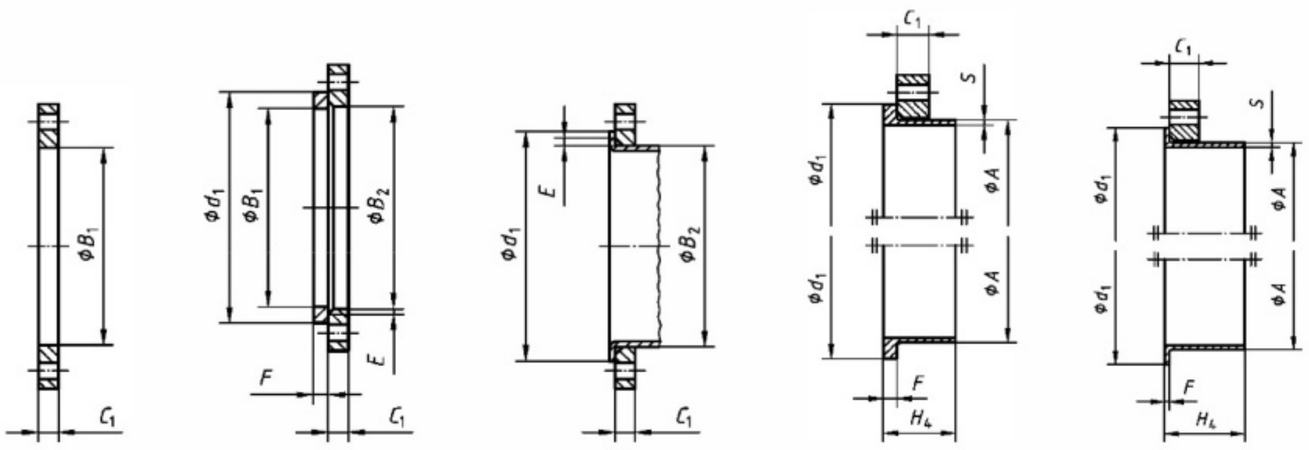
NOTE 1 Dimensions N_1, N_2 and N_3 are measured at the intersection of the hub draft angle and the back face of the flange.

NOTE 2 Type 33; lapped pipe end without determination of thickness and height.

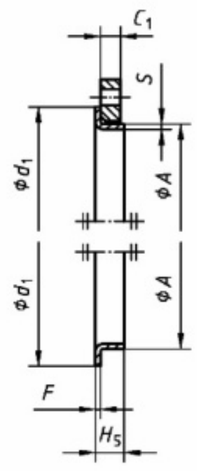
DIMENSIONS OF PN 16 FLANGES



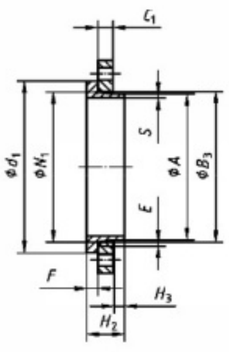
This diagram illustrates the arrangement but not necessarily the correct number of bolt holes. Refer to the column "Bolting Number" for the actual number.



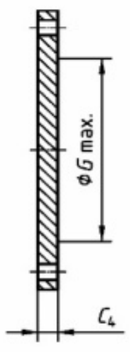
Type 01 | Type 02 and 32 | Type 02 and 33 | Type 02 and 35 | Type 02 and 36



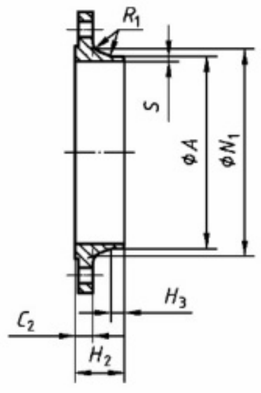
Type 02 and 37



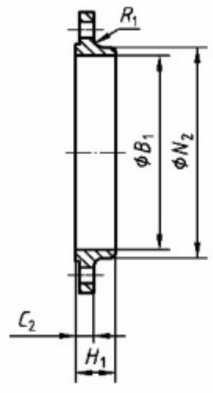
Type 04 and 34



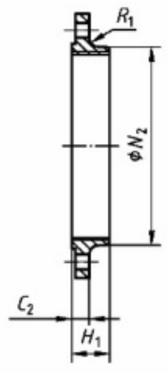
Type 05



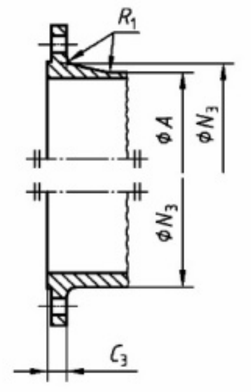
Type 11



Type 12



Type 13



Type 21

DN	Mating dimensions					Outside diameter of neck	Bore diameters			Flange thickness				Chamfer	Collar thickness			Diameter of shoulder	Length					Neck diameters			Corner radii	
	Outside diameter	Diameter of bolt circle	Diameter of bolt hole	Bolting																								
				Number	Size																							
	D	K	L			A	B_1	B_2	B_3	C_1	C_2	C_3	C_4	E	F			G_{max}	H_1	H_2	H_3	H_4	H_5	N_1	N_2	N_3	R_1	
	Flange type																											
	01, 02, 04, 05, 11, 12, 13, 21					11 21 [a] 34 [d] 35 - 37	01 12 32	02	04	01 02 04	11 12 13	21	05	02 04	32 34	35	36	37	05	12 13	11 34 [c]	11 34 c	35	36	37	11 34 [c]	12 13	21	11 12 13 21, 34	34
10	90	60	14	4	M12	17,2	18,0	21	31	14	16	16	16	3	12	5	2	2,5	—	22	35	6	35	35	7	28	30	28	4	1,8
15	95	65	14	4	M12	21,3	22,0	25	35	14	16	16	16	3	12	5	2	2,5	—	22	38	6	38	38	7	32	35	32	4	2,0
20	105	75	14	4	M12	26,9	27,5	31	42	16	18	18	18	4	14	6	2,5	3	—	26	40	6	40	40	8	40	45	40	4	2,3
25	115	85	14	4	M12	33,7	34,5	38	49	16	18	18	18	4	14	7	2,5	3	—	28	40	6	40	40	10	46	52	50	4	2,6
32	140	100	18	4	M16	42,4	43,5	47	59	18	18	18	18	5	14	8	3	3	—	30	42	6	42	42	12	56	60	60	6	2,6
40	150	110	18	4	M16	48,3	49,5	53	67	18	18	18	18	5	14	8	3	3	—	32	45	7	45	45	15	64	70	70	6	2,6
50	165	125	18	4	M16	60,3	61,5	65	77	20	18	18	18	5	16	8	3	4	—	28	45	8	45	45	20	74	84	84	6	2,9
65	185	145	18	8 [b]	M16	76,1	77,5	81	96	20	18	18	18	6	16	8	3	4	55	32	45	10	45	45	20	92	104	104	6	2,9
80	200	160	18	8	M16	88,9	90,5	94	108	20	20	20	20	6	16	10	3	4	70	34	50	10	50	50	25	105	118	120	6	3,2
100	220	180	18	8	M16	114,3	116,0	120	134	22	20	20	20	6	18	10	4	4	90	40	52	12	52	52	25	131	140	140	8	3,6
125	250	210	18	8	M16	139,7	141,5	145	162	22	22	22	22	6	18	10	4	4	115	44	55	12	55	55	25	156	168	170	8	4,0
150	285	240	22	8	M20	168,3	170,5	174	188	24	22	22	22	6	20	10	5	5	140	44	55	12	55	55	25	184	195	190	10	4,5
200	340	295	22	12	M20	219,1	221,5	226	240	26	24	24	24	6	20	11	6	6	190	44	62	16	62	62	30	235	246	246	10	6,3
250	405	355	26	12	M24	273,0	276,5	281	294	29	26	26	26	8	22	12	10	—	235	46	70	16	70	68	—	292	298	296	12	6,3
300	460	410	26	12	M24	323,9	327,5	333	348	32	28	28	28	8	24	14	10	—	285	46	78	16	78	68	—	344	350	350	12	7,1
350	520	470	26	16	M24	355,6	359,0	365	400	35	30	30	30	8	26	18	10	—	330	57	82	16	82	68	—	390	400	410	12	8,0
400	580	525	30	16	M27	406,4	411,0	416	454	38	32	32	32	8	28	20	10	—	380	63	85	16	85	72	—	445	456	458	12	8,0
450	640	585	30	20	M27	457,0	462,0	467	500	42	34	40	40	8	30	22	—	—	425	68	83	16	87	—	—	490	502	516	12	8,0
500	715	650	33	20	M30	508,0	513,5	519	556	46	36	44	44	8	32	22	—	—	475	73	84	16	90	—	—	548	559	576	12	8,0
600	840	770	36	20	M33	610,0	616,5	622	660	55	40	54	54	8	32	24	—	—	575	83	88	18	95	—	—	670	658	690	12	8,8
700	910	840	36	24	M33	711,0		721	—	63	40		58	8	—	26	—	—	670	83	104	18	100	—	—	755	760	760	12	—
800	1 025	950	39	24	M36	813,0	c	824	—	74	41		62	8	—	28	—	—	770	90	108	20	105	—	—	855	864	862	12	—

[a] For flanges type 21 the outside hub diameter approximately corresponds to the outside pipe diameter.
[b] According to EN 1092-2 (Cast iron flanges) and EN 1092-3 (Copper alloy flanges), the flanges in this DN and PN may be supplied with 4 holes. Where steel flanges are required with 4 holes, these may be supplied by agreement between flange manufacturer and purchaser.
[c] To be specified by the purchaser.
[d] Use is limited up to DN 600.

NOTE 1 Dimensions N_1, N_2 and N_3 are measured at the intersection of the hub draft angle and the back face of the flange.
NOTE 2 Type 33; lapped pipe end without determination of thickness and height.

DIMENSIONS OF PLATE FLANGES AS PER TABLE BS-10

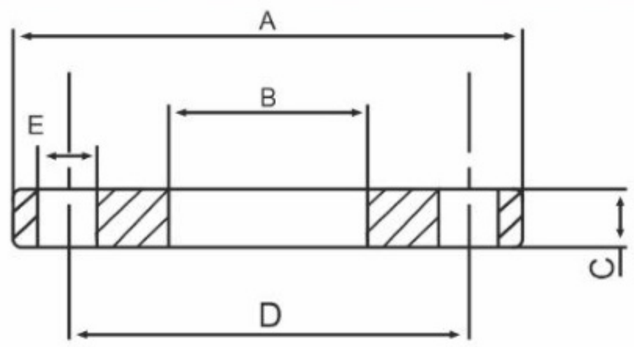


TABLE - D

For Working Stream Pressure above 250 lbs and upto 350 lbs per sq. inch

Nominal Pipe Size (in mm)	Dia of Flange	Dia of Bolt Circle (PCD)	No of Bolt	Dia of Bolt	Thickness of Flange
mm	mm	mm	mm	mm	mm
15	95	67	4	13	5
20	102	73	4	13	5
25	114	83	4	13	5
32	121	87	4	13	6
40	133	98	4	13	6
50	152	114	4	16	8
65	165	127	4	16	8
80	184	145	4	16	10
100	216	178	4	16	10
125	254	210	8	16	13
150	279	235	8	16	13
200	337	292	8	16	13
250	406	356	8	19	16
300	457	406	12	19	16
350	527	470	12	22	19
400	578	521	12	22	19
450	641	584	12	22	22
500	705	641	16	22	25
600	826	756	16	25	29

TABLE - E

For Working Stream Pressure above 350 lbs and upto 450 lbs per sq. inch

Nominal Pipe Size (in mm)	Dia of Flange	Dia of Bolt Circle (PCD)	No of Bolt	Dia of Bolt	Thickness of Flange
mm	mm	mm	mm	mm	mm
16	95	67	4	13	6
20	102	73	4	13	6
25	114	83	4	13	7
32	121	87	4	13	8
40	133	98	4	13	9
50	152	114	4	16	10
65	165	127	4	16	10
80	184	145	4	16	11
100	216	178	8	16	13
125	254	210	8	16	14
150	279	235	8	19	17
200	337	292	8	19	19
250	406	356	12	19	22
300	457	406	12	22	25
350	527	470	12	22	25
400	578	521	12	22	25
450	641	584	16	22	29
500	705	641	16	22	32
600	826	756	16	25	38

TABLE - F

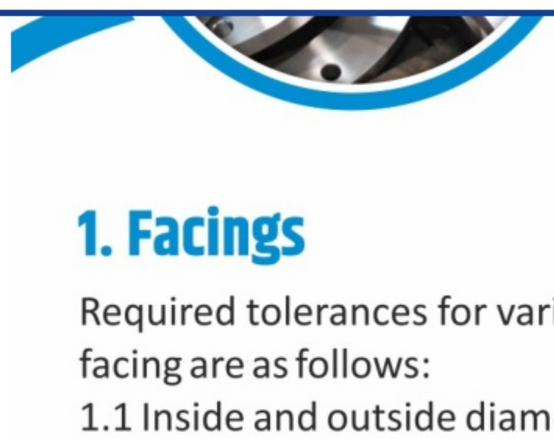
For Working Stream Pressure above 250 lbs and upto 350lbs per sq. inch

Nominal Pipe Size (in mm)	Dia of Flange	Dia of Bolt Circle (PCD)	No of Bolt	Dia of Bolt	Thickness of Flange
mm	mm	mm	mm	mm	mm
15	95	67	4	13	10
20	102	73	4	13	10
25	121	87	4	16	10
32	133	98	4	16	13
40	140	105	4	16	13
50	165	127	4	16	16
65	184	145	8	16	16
80	203	165	8	16	16
100	229	191	8	16	19
125	279	235	8	19	22
150	305	260	12	19	22
200	368	324	12	14	25
250	432	381	12	22	25
300	489	438	16	25	29
350	552	495	16	25	32
400	610	552	20	25	32
450	673	610	20	29	35
500	737	673	24	29	38
600	851	781	24	32	41

TABLE - H

For Working Stream Pressure above 350 lbs and upto 450 lbs per sq. inch

Nominal Pipe Size (in mm)	Dia of Flange	Dia of Bolt Circle (PCD)	No of Bolt	Dia of Bolt	Thickness of Flange
mm	mm	mm	mm	mm	mm
15	114	83	4	16	13
20	114	83	4	16	13
25	121	87	4	16	14
32	133	98	4	16	17
40	140	105	4	16	17
50	165	127	4	16	19
65	184	145	8	16	19
80	203	165	8	16	22
100	229	191	8	16	25
125	279	235	8	19	29
150	305	260	12	19	29
200	368	324	12	19	32
250	432	381	12	22	35
300	489	438	16	22	38
350	552	495	16	25	41
400	610	552	20	25	44
450	673	610	20	29	48
500	737	673	24	29	51
600	851	781	24	32	57



GENERAL TOLERANCE

1. Facings

Required tolerances for various flange and flanged fitting facing are as follows:

1.1 Inside and outside diameter of large and small tongue and Groove and female, ± 0.5mm
1.2 Outside diameter, 2.0 mm Raised face, ± 1.0 mm
1.3 Outside diameter, 7.0 mm Raised face, ± 0.5 mm

2. Flange Thickness

Required tolerances for flanges thickness are as follows.

NPS ≤ 18	+3.0, -0.0 mm
NPS ≥ 20	+5.0, -0.0 mm

The plus tolerances is applicable to bolting bearing surfaces whether as-forged, as-cast, spot-faced or back-faced.

3. Welding End Flange Ends and Hubs

3.1 Outside Diameter : Required tolerances for the nominal outside diameter dimension A of figs, welding end of welding neck flanges are as follow:

NPS ≤ 5	+2.0, -1.0 mm
NPS ≥ 6	+ 4.0, -1.0 mm

3.2 Inside Diameter : Required tolerances for the nominal inside diameter of welding ends of welding neck flanges and smaller bore of socket welding neck flanges and smaller bore of socket welding langes (dimension B in the referenced figures) are as follows.

NPS ≤ 10	± 1.0 mm
12 ≤ NPS <= 18	± 1,5 mm
NPS ≥ 20,	+3.0, -1.5MM

3.3 Hub Thickness : Despite the tolerances specified for dimensions A and B, the thickness of the hub at the welding end shall not be less than 87 1/2% of the nominal thickness. of the pipe, having an under tolerance of 12,5% for the pipe wall thickness to which the flanges is to be attached, or the minimum wall thickness as specified by the purchaser.



4. Length Through Hub on Welding Neck Flanges

The required tolerances for the length through hubs on welding neck flanges are as follows.

NPS ≤ 4	± 1.5 mm
5 < NPS ≤ 10	+ 1.5, -3.0 mm
NPS ≥ 12	+3.0mm-5.0 mm

5. Flange Bore

5.1 Lapped, Slip-on, and Socket Welding Flanges Bores. The required tolerances for lapped, slip-on, and socket welding flanges bores are as follows.

NPS ≤ 10	+ 1.0, 0.0 mm
NPS ≥ 12	+1.5, -0.0 mm

5.2 Counterbores, Threaded Flanges. The required tolerances for the threaded flanges counterbores are follows.

NPS ≤ 10	+1.0, -0.0 mm
NPS ≥ 12	+1.5, -0.0 mm

5.3 Counterbores, Socket Welding Flanges.
The required tolerances for socket end counterbores is as follows.

1/2 < NPS < 3	+ 0.25mm

6. Drilling and Facing

6.1 Bolt Circle Diameter : The required tolerance for all bolt circle diameters is +1.5 mm

6.2 Bolt Hole to Bolt Hole

The required tolerance for the center-to-center of adjacent bolt holes + 0.8 mm

6.3 Bolt Circle Concentricity

The required tolerances for concentricity between the flanges diameters are as follows.

NPS ≤ 2 1/2"	0.8 mm
NPS ≥ 3"	1.5 mm